ALLEGRETTI GIOVANNI

Senior C++ Software Engineer

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Giovanni Allegretti g-allegretti Portfolio

01. SUMMARY

Senior C++ Software Engineer with six years of experience in performance-critical, real-time systems: multithreaded pipeline architecture, GPU inference deployment and optimisation (CUDA, TensorRT), computer vision, and Qt/QML desktop applications. Strong background in software architecture design and CI/CD integration.

02. EXPERIENCE

Senior Software Engineer

Apr 2023 – Present

DMA (Plasser Diagnostics, Plasser & Theurer Group) Turin (IT) - hybrid, fully remote since 2025

- Lead the architecture of the real-time video inspection framework, now being brought to high-speed train inspection: redesigned it from a thread-per-stage pipeline into a DAG-based execution model (Taskflow) with GPU-aware scheduling and decode/analysis overlap, delivering roughly 2-3x per-GPU inference throughput.
- Designed the GPU-accelerated inference layer (TensorRT) and scaled it across up to 4 GPUs ($\sim$4.6x throughput), sustaining $\sim$99% GPU utilisation where the previous serial-decode design dropped to $\sim$40%. Validated on multi-GPU hardware (4x RTX 3090, NVIDIA L40S).
- Led the full redesign of the catenary wear measurement product: ground-up Qt/QML UI rewrite, generalisation of the analysis from 1 to up to 8 wires, rebuild of the multi-source data synchronisation, and faster processing. Currently extending it to line-scan cameras running at up to 5 kHz for high-speed train inspection.
- Designed and built a standalone Qt/QML real-time monitoring application around the inspection framework: live camera streams, defect detection results and runtime benchmarks.
- Containerised the core libraries, packaged dependencies into a self-hosted Debian/APT repository, and built multi-platform CI/CD pipelines (Linux GCC + Windows MSVC).
- Review colleagues' code and support them on design and implementation choices, raising code quality across the team.

Software Developer

Jul 2020 – Apr 2023

ALTEN - consultant at Leonardo Aircraft Division Turin (IT)

- Designed and developed HMI applications in C++ and Qt for avionics mission simulation, enabling real-time monitoring of cockpit inputs and injection of simulated failures.
- Developed and maintained a flight and mission simulation model in C++, focusing on reliability, performance, and system integration.

Software Engineer (Internship)

Mar 2019 – Jul 2019

ALTEC - Aerospace Logistics Technology Engineering Company Turin (IT)

- Designed, developed, and tested a C++ software module for planetary exploration robots, computing terrain-adaptive kinematics for a 6-DOF robotic arm, with roll/pitch error $\leq$ 0.10 rad and Z error $\leq$ 0.10 m on inclined slopes. Integrated into an existing ROS/Gazebo framework.

03. SKILLS

Languages	Modern C++ (C++17, C++20), CUDA C++, Python, C, QML, Bash, MATLAB
Core	Software Architecture, Multithreading, Real-Time Systems, Computer Vision, Design Patterns
Methodologies	Agile/Scrum
GPU & Inference	CUDA, TensorRT, DAG task scheduling (Taskflow), NVIDIA Jetson
Libraries & Frameworks	Qt 5/6, Qt Quick/QML, OpenCV, Boost, ZMQ, GTest, ROS, Gazebo, TensorFlow
Build, CI/CD & Tools	CMake, vcpkg, Docker, Debian Packaging, GitLab CI/CD, Git, GitHub, gdb, Valgrind, Intel VTune, Kestra
Platforms	Linux (GCC), Windows (MSVC), Embedded (STM32, ESP32)
Spoken Languages	Italian (native), English (advanced, professional working proficiency)
Personal Traits	Problem Solving, Teamwork, Critical Thinking, Attention to Detail, Continuous Learning

04. CERTIFICATIONS

Getting Started with Accelerated Computing in Modern CUDA C++ - NVIDIA Deep Learning Institute Credential · Oct 2026

05. EDUCATION

Facial Expression Recognition - Computer Vision
2019

Master Thesis, Erasmus Program - **University of Groningen** Groningen (NL)

- Automatic Facial Expression Recognition in unconstrained ("in-the-wild") video, using temporal modelling (CNN+LSTM, C3D) and multimodal video+audio fusion.

Master's Degree in Computer Engineering - 110/110
2020

Università degli Studi di Salerno Fisciano (IT)

Bachelor's Degree in Computer Engineering - 95/110
2017

Università degli Studi di Salerno Fisciano (IT)

06. PROJECTS AND COLLABORATIONS

ROS Object Grasping - Cognitive Robotics
2019

Università degli Studi di Salerno Fisciano (IT)

- Designed and developed a cognitive robotics system for object recognition and manipulation using computer vision and neural networks.

Autonomous Vision Rover - Computer Vision
2019

Università degli Studi di Salerno Fisciano (IT)

- Developed an autonomous rover system for navigation in a controlled environment.

STM32 Differential-Drive Rover - Embedded Systems
2018

Università degli Studi di Salerno Fisciano (IT)

- Designed and implemented a custom PID closed-loop controller and firmware for both manual and autonomous (line-following) driving modes on an STM32F401RETx.